AD Fake Account Detection System

Portfolio report for a Python-based Active Directory fake account detection project.

Component	Description
Language	Python 3
Input	Windows Security logs exported as JSON/JSONL
Output	HTML report, CSV summary, JSON findings
Focus	Fake/suspicious Active Directory account detection
Status	Working prototype with generated sample outputs

Project Summary

The detector analyzes exported Windows Security events, normalizes log fields, correlates account-centered timelines, applies weighted risk scoring, and generates analyst-readable investigation reports.

Detection Signals

- Account creation outside business hours or on weekends
- Suspicious service-like, privileged, or random-looking usernames
- Account enablement shortly after creation
- Fast membership in privileged security groups
- Kerberos TGT/TGS activity after suspicious creation
- Failed and successful logon activity connected to the account timeline
- Password reset/change and directory object modification events

Included Output Runs

Scenario	Parsed Events	Findings	HIGH	MEDIUM	Top Accounts
Baseline medium	192	3	0	3	root, user1, suspectuser
Advanced high risk	84	7	6	1	random123abc456def, svc_backup, admin, guest, support

Repository Layout

```
src/ad_fake_account_detector.py - detector source code
outputs/ - generated findings, summaries, and HTML reports
docs/ - diploma PDF, event ID reference, detection logic, and this portfolio report
examples/ - example workflow notes
```

Safety Scope

The project is a defensive log analysis tool. It does not authenticate to systems, modify Active Directory, perform exploitation, or run privilege escalation. Public datasets and screenshots should remain anonymized.